

HOSSAM NAWAR

COMPUTER SCIENCE | ROBOTICS | STEM EDUCATOR

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SUMMARY

Computer Science, Robotics, and STEM Educator with over 6 years of experience inspiring students through project-based learning and hands-on technology education. Experienced in teaching Python, HTML, CSS, JavaScript, Scratch, Robotics, Arduino, IoT, LEGO Spike Prime, and EV3 across different age groups. Passionate about developing students' creativity, computational thinking, collaboration, and problem-solving skills through engaging classroom experiences and real-world projects. My background in Architectural Engineering has strengthened my abilities in organization, structured thinking, and creative design, enabling me to create well-planned learning experiences that prepare students for future careers in technology.

PROFESSIONAL EXPERIENCE

- **Senior STEM, Robotics & Programming Instructor** **Future Technology Academy**
 - **Jun 2021 – Present**
 - Designed and delivered STEM, Computer Science, Coding, and Robotics programs for students across different age groups.
 - Taught Python, HTML, CSS, JavaScript, Scratch, Pictoblox, MIT App Inventor, Arduino, LEGO Spike Prime, EV3 Mindstorms, LEGO WeDo, and LEGO Boost.
 - Designed project-based learning experiences that enhanced creativity, computational thinking, problem-solving, teamwork, and presentation skills.
 - Developed educational curricula, lesson plans, classroom activities, and hands-on coding projects aligned with modern teaching methodologies.
 - Mentored student teams for national and international coding and robotics competitions, achieving multiple award-winning results.
 - Assessed students' progress, provided constructive feedback, and supported their academic and personal development.
 - Created an engaging classroom environment that encouraged innovation, critical thinking, and confidence in using technology.
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OTHER PROFESSIONAL EXPERIENCE

Microfinance Risk Management Engineer

Future Business Women Association

- **Jan 2023 – Present**
 - Conduct financial risk assessments and analyze financing applications to support informed decision-making.
 - Review clients' financial records and ensure compliance with organizational policies and risk standards.
 - Prepare risk reports, monitor key risk indicators, and recommend appropriate risk mitigation strategies.
 - Develop strong analytical, organizational, and decision-making skills that complement my structured approach to education.
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CORE COMPETENCIES

- Computer Science Education
 - Robotics Education
 - STEM Education
 - Coding Curriculum Design
 - Project-Based Learning
 - Educational Technology Integration
 - Competition Preparation & Coaching
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SKILLS

TECHNICAL SKILLS

- PYTHON
- HTML & CSS
- JAVASCRIPT
- SCRATCH
- PICTOBLOX
- MIT APP INVENTOR
- ARDUINO
- FIGMA
- FLUTTER (BASIC)
- DART (BASIC)

ROBOTICS & STEM

- LEGO SPIKE PRIME
- LEGO MINDSTORMS EV3
- LEGO WEDO
- LEGO BOOST
- ROBOTICS PROGRAMMING
- SENSOR INTEGRATION
- STEM EDUCATION
- ENGINEERING DESIGN PROCESS

DIGITAL TOOLS

- MICROSOFT OFFICE
- GOOGLE WORKSPACE
- CANVA
- ZOOM
- GOOGLE CLASSROOM

TEACHING & EDUCATIONAL SKILLS

- CURRICULUM DEVELOPMENT
- PROJECT-BASED LEARNING (PBL)
- LESSON PLANNING
- CLASSROOM MANAGEMENT
- STUDENT ASSESSMENT
- DIFFERENTIATED INSTRUCTION
- CODING & ROBOTICS COMPETITION COACHING
- STUDENT MENTORING
- PRESENTATION & PUBLIC SPEAKING

PROFESSIONAL SKILLS

- COMMUNICATION
 - LEADERSHIP
 - TEAMWORK
 - PROBLEM SOLVING
 - CRITICAL THINKING
 - TIME MANAGEMENT
 - ORGANIZATION
 - CREATIVITY
 - ADAPTABILITY
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EDUCATION

Bachelor of Architectural Engineering

Sep 2015 - Sep 2020

- Faculty Of Engineering (El-Sherouk Academy)

Strong foundation in architectural design, analytical thinking, and creative project development

Diploma in Project Management Engineering

Jan 2021

- Kafr Elsheikh University Degree : Very Good

Professional Diploma in Building Information Modeling

Jan 2023

- British University Degree : Good
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LANGUAGES

ARABIC — NATIVE

ENGLISH — PROFESSIONAL WORKING PROFICIENCY

COMPETITION COACHING & STUDENT ACHIEVEMENTS

ROBOTICS & PROGRAMMING COACH (2021 – 2026)

- **AFRO-ASIAN FORUM FOR INNOVATION & TECHNOLOGY (2021)**

COACHED STUDENTS TO ACHIEVE 4TH PLACE IN THE SCIENCE & ROBOTICS CARNIVAL.

- **AFRO-ASIAN FORUM FOR INNOVATION & TECHNOLOGY FINALS – DUBAI (2022)**

MENTORED STUDENT TEAMS TO ACHIEVE 1ST PLACE IN THE SCIENCE & ROBOTICS CARNIVAL AND 1ST PLACE IN THE BOWLING ROBOT COMPETITION.

- **AFRO-ASIAN FORUM FOR INNOVATION & TECHNOLOGY (2023)**

GUIDED STUDENTS TO ACHIEVE 2ND PLACE IN THE SCIENCE & ROBOTICS CARNIVAL AND 2ND PLACE IN THE SOFTWARE LEADER (JUNIOR) COMPETITION.

- **AFRO-ASIAN FORUM FOR INNOVATION & TECHNOLOGY FINALS - SAUDI ARABIA**

GUIDED STUDENTS TO ACHIEVE 1ST PLACE IN THE SCIENCE & ROBOTICS CARNIVAL AND 2ND PLACE IN THE SOFTWARE LEADER (SENIOR) COMPETITION.

- **FIRST LEGO LEAGUE (2023)**

MENTORED TEAMS THAT RECEIVED THE BREAKTHROUGH AWARD AND THE BEST MODEL AWARD.

- **FIRST LEGO LEAGUE CHALLENGE (2024)**

COACHED STUDENTS TO ACHIEVE 3RD PLACE – ROBOT PERFORMANCE.

- **ROBOTEX INTERNATIONAL (2024)**

COACHED STUDENTS TO ACHIEVE 1ST PLACE IN THE FOLLOW LINE ENHANCED COMPETITION.

- **AFRO-ASIAN FORUM FOR INNOVATION & TECHNOLOGY (2026)**

SUPERVISED STUDENT TEAMS THAT ACHIEVED:

1ST PLACE – MOBILE APPLICATION (SENIOR)

2ND PLACE – MOBILE APPLICATION (JUNIOR)

3RD PLACE – FLAG COLLECTOR ROBOT COMPETITION

SUMMARY OF ACHIEVEMENTS

SUCCESSFULLY COACHED AND MENTORED STUDENT TEAMS IN NATIONAL AND INTERNATIONAL ROBOTICS, CODING, STEM, AND INNOVATION COMPETITIONS, LEADING THEM TO ACHIEVE MULTIPLE TOP-RANKING AWARDS BETWEEN 2021 AND 2026.
